

A PAIRWISE LEARNING-TO-RANK MODEL FOR FEATURE ENGINEERING, LOGISTIC OBJECTIVES, AND CALIBRATION

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ABSTRACT. We develop an interpretable learning-to-rank pipeline for first-round admissions screening in Pi Mu Epsilon, using limited human pairwise comparisons [1, 2, 3] within the pairwise learning-to-rank framework [4, 5, 6, 7]. A deterministic map $x : \mathcal{A} \rightarrow [0, 10]^{21}$ encodes structured fields, resume text, and short-answer signals as bounded named features [8, Chapter 3]. The linear score $s_w(a) := \langle w, x(a) \rangle$ minimizes the regularized logistic risk [5, 8]

$$L_\lambda(w) := \frac{1}{|\mathcal{D}|} \sum_{(i,j,y_{ij}) \in \mathcal{D}} \log(1 + \exp(-z_{ij} \langle w, x(a_i) - x(a_j) \rangle)) + \frac{\lambda}{2} \|w\|_2^2, \quad z_{ij} := 2y_{ij} - 1.$$

For $\lambda > 0$, this objective is λ -strongly convex and $(\lambda + 25 \cdot 21)$ -smooth; hence gradient descent converges linearly to its unique minimizer [9, Chapter 3]. Raw scores are calibrated by

$$\hat{s}(a) := 10 \hat{F}_w(s_w(a)),$$

which maps them to $[0, 10]$, preserves order, and is invariant under strictly increasing reparametrization [7], [8, Chapter 4]. Finally, we specify the reproducible data artifacts `applicants.jsonl`, `pairs.csv`, and `rank_model.json` linking local training to deployed scoring.

CONTENTS

Part 1. Pairwise Ranking	2
1. Introduction	2
1.1. Setting and result	2
2. Preliminaries	2
2.1. Ranking models, the logistic function, and linear bounds	3
3. From applications to vectors	4
3.1. The five feature blocks	4
4. Linear scoring model	5
4.1. Scores, level sets, and scale	6
5. Pairwise training objective	6
5.1. Loss, gradient, convexity, and gradient descent	7
6. Calibration and reporting	8
6.1. Percentile calibration and decomposed reporting	9
7. Proof of Theorem A	9
8. The conjecture	10
8.1. The regularization path and its limit	10
Part 2. Appendix	12
Appendix A. Foundational tools	12
A.1. Quadratic bounds and maximum likelihood	12
Appendix B. Data artifacts	12
B.1. Records, pairs, and the model file	12

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Acknowledgments	13
References	13

Part 1. Pairwise Ranking

1. INTRODUCTION

First-round admissions screening produces a consistent and auditable ordering that allocates reviewer attention rather than a final decision, and in the regime of $n \approx 10\text{--}50$ applicants per cycle the only reliably collectable supervision is the comparison $a_i \succ a_j$ between two applications [1], [4], [5], [7]. Throughout, $\mathcal{A}$ is the space of applications, $d \in \mathbb{Z}_{\geq 1}$ the feature dimension, $\langle \cdot, \cdot \rangle$ the Euclidean inner product on $\mathbb{R}^d$ with norm $\|\cdot\|_2$, $\mathbf{1}\{\cdot\}$ an indicator, I the identity matrix, $\preceq$ the Loewner order on symmetric matrices, and

$$\sigma(t) := \frac{1}{1 + e^{-t}} \quad (t \in \mathbb{R}), \quad \sigma(-t) = 1 - \sigma(t), \quad \sigma'(t) = \sigma(t) \sigma(-t).$$

1.1. Setting and result.

Definition 1.1.1 (Features, scores, and margins). A feature map is a deterministic function $x: \mathcal{A} \rightarrow [0, 10]^d$, $a \mapsto x(a) = (x_1(a), \dots, x_d(a))$, carrying a schema identifier $\text{fv} \in \mathbb{Z}_{\geq 1}$. For $w \in \mathbb{R}^d$ and applications a, a_i, a_j ,

$$s_w(a) := \langle w, x(a) \rangle = \sum_{k=1}^d w_k x_k(a), \quad d_{ij} := x(a_i) - x(a_j), \quad \Delta_{ij}(w) := s_w(a_i) - s_w(a_j) = \langle w, d_{ij} \rangle.$$

Definition 1.1.2 (Pairwise dataset and regularized pairwise logistic risk, [1, 5]). A pairwise dataset is a finite multiset $\mathcal{D} = \{(i, j, y_{ij})\}$ of pairs of applicant indices with labels $y_{ij} \in \{0, 1\}$, $y_{ij} = 1$ meaning $a_i \succ a_j$, and $z_{ij} := 2y_{ij} - 1 \in \{-1, +1\}$. The Bradley–Terry probability, the pairwise logistic loss, and the ℓ_2 -regularized risk are

$$\begin{aligned} \mathbb{P}(a_i \succ a_j \mid w) &:= \sigma(\Delta_{ij}(w)), \quad \ell_{ij}(w) := \log(1 + \exp(-z_{ij} \Delta_{ij}(w))), \\ L_\lambda(w) &:= \frac{1}{|\mathcal{D}|} \sum_{(i,j,y) \in \mathcal{D}} \ell_{ij}(w) + \frac{\lambda}{2} \|w\|_2^2 \quad (\lambda \geq 0), \quad \Lambda := \lambda + \frac{1}{4|\mathcal{D}|} \sum_{(i,j,y) \in \mathcal{D}} \|d_{ij}\|_2^2. \end{aligned}$$

Theorem A (Regularized pairwise logistic ranking). *Let $x: \mathcal{A} \rightarrow [0, 10]^d$ be a feature map, $\mathcal{D}$ a pairwise dataset, $\lambda > 0$, $w^{(t+1)} := w^{(t)} - \Lambda^{-1} \nabla L_\lambda(w^{(t)})$, and $\hat{s}$ the calibrated score of Definition 6.1.1 on a pool $\mathcal{P}$. Then L_λ has a unique minimizer w_λ , $\Lambda \leq \lambda + 25d$, and for all $w \in \mathbb{R}^d$, $t \geq 0$, and $a, b \in \mathcal{P}$,*

$$\begin{aligned} \lambda I &\preceq \nabla^2 L_\lambda(w) \preceq \Lambda I, \quad s_w(a) \leq s_w(b) \iff \hat{s}(a) \leq \hat{s}(b), \\ L_\lambda(w^{(t)}) - L_\lambda(w_\lambda) &\leq \left(1 - \frac{\lambda}{\Lambda}\right)^t (L_\lambda(w^{(0)}) - L_\lambda(w_\lambda)). \end{aligned}$$

Proof. §7 □

2. PRELIMINARIES

2.1. Ranking models, the logistic function, and linear bounds.

Definition 2.1.1 (Bradley–Terry and Plackett–Luce models, [1, 2, 3]). For $\theta \in \mathbb{R}^n$ and a permutation π of $\{1, \dots, n\}$,

$$\mathbb{P}_\theta(i \succ j) := \frac{e^{\theta_i}}{e^{\theta_i} + e^{\theta_j}} = \sigma(\theta_i - \theta_j), \quad \mathbb{P}_\theta(\pi) := \prod_{k=1}^n \frac{e^{\theta_{\pi(k)}}}{\sum_{l=k}^n e^{\theta_{\pi(l)}}},$$

the second reducing to the first for $n = 2$.

Fact 2.1.2 (Logistic identities, [8]). For $t \in \mathbb{R}$,

$$\begin{aligned} \sigma(-t) &= 1 - \sigma(t), & \sigma'(t) &= \sigma(t) \sigma(-t) \in (0, \tfrac{1}{4}], & -\log \sigma(t) &= \log(1 + e^{-t}), \\ \frac{d}{dt} \log(1 + e^t) &= \sigma(t), & \frac{d^2}{dt^2} \log(1 + e^t) &= \sigma'(t) > 0, & \frac{d^2}{dt^2} \log(1 + e^{-t}) &= \sigma'(-t) > 0. \end{aligned}$$

Lemma 2.1.3 (Linear functionals on a box). For $w, v \in \mathbb{R}^d$ with $\|v\|_\infty \leq 10$,

$$|\langle w, v \rangle| \leq \|w\|_1 \|v\|_\infty \leq 10 \|w\|_1, \quad \|v\|_2^2 \leq d \|v\|_\infty^2 \leq 100d, \quad v, w \in [0, \infty)^d \implies 0 \leq \langle w, v \rangle \leq 10 \|w\|_1.$$

Proof.

$$|\langle w, v \rangle| \leq \sum_{k=1}^d |w_k| |v_k| \leq \|v\|_\infty \|w\|_1, \quad \|v\|_2^2 = \sum_{k=1}^d v_k^2 \leq d \|v\|_\infty^2, \quad v_k, w_k \geq 0 \implies 0 \leq \langle w, v \rangle \leq 10 \|w\|_1.$$

□

Lemma 2.1.4 (Level sets of a linear functional). For $w \in \mathbb{R}^d \setminus \{0\}$, $\hat{w} := w/\|w\|_2$, $w^\perp := \{\eta : \langle w, \eta \rangle = 0\}$, and $c, c' \in \mathbb{R}$,

$$H_w(c) := \{\xi \in \mathbb{R}^d : \langle w, \xi \rangle = c\} = \frac{c}{\|w\|_2} \hat{w} + w^\perp, \quad \text{dist}(H_w(c), H_w(c')) = \frac{|c - c'|}{\|w\|_2}.$$

Proof.

$$\begin{aligned} \langle w, \xi \rangle = c &\iff \xi - \frac{c}{\|w\|_2} \hat{w} \in w^\perp, & \xi \in H_w(c), \eta \in H_w(c') &\implies \|\xi - \eta\|_2 \geq |\langle \hat{w}, \xi - \eta \rangle| = \frac{|c - c'|}{\|w\|_2}, \\ \eta &:= \xi + \frac{c' - c}{\|w\|_2} \hat{w} \in H_w(c') &\implies \|\xi - \eta\|_2 &= \frac{|c - c'|}{\|w\|_2}. \end{aligned}$$

□

Remark 2.1.5 (Linear Bradley–Terry models). Definition 1.1.2 is Definition 2.1.1 with $\theta_i = s_w(a_i)$, and Bradley–Terry on n items is the case $d = n$, $x(a_i) = e_i$, $w = \theta$,

$$\mathbb{P}(a_i \succ a_j \mid w) = \sigma(s_w(a_i) - s_w(a_j)) = \mathbb{P}_{(s_w(a_m))_m}(i \succ j), \quad x(a_i) = e_i \implies \Delta_{ij}(w) = w_i - w_j.$$

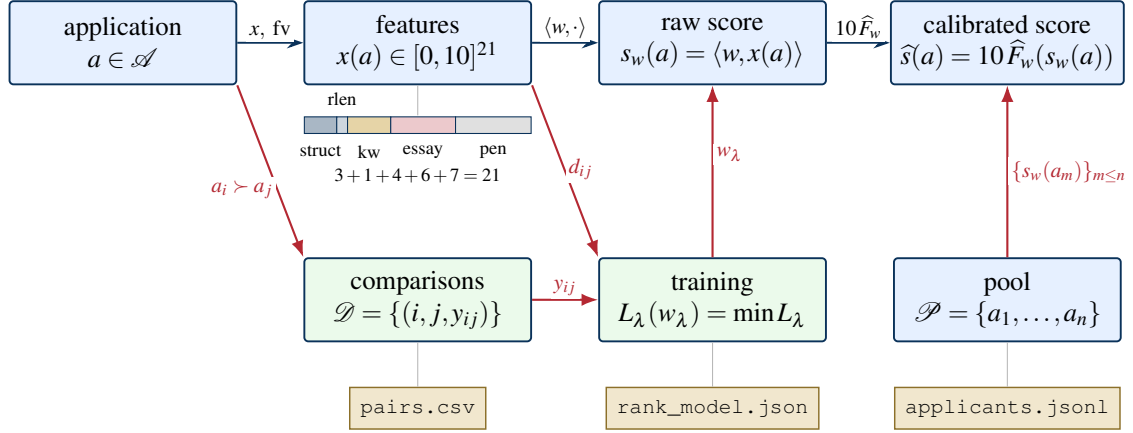


FIGURE 1. The pipeline. Applications are mapped deterministically to 21 bounded coordinates in five blocks, scored linearly, and calibrated against the pool, while human comparisons and feature differences train w_λ ; the three files record labels, weights, and pool records.

3. FROM APPLICATIONS TO VECTORS

Every coordinate of the feature map is a bounded deterministic functional of one signal, so all randomness enters through the training of § 5 and none through extraction [5], [8, Chapter 3]. Throughout, Σ^* is the set of finite strings, $v \sqsubseteq T$ means that the phrase v occurs in T , $d = 21$, and an application a carries its grade point average, calculus completion, entered course codes, browser-extracted resume text, and two short answers,

$$\text{GPA}(a) \in [0, 4.33], \quad \text{Calc}(a) \in \{\text{yes}, \text{no}\}, \quad U(a) \subset \Sigma^*, \quad T(a), E_A(a), E_B(a) \in \Sigma^*.$$

3.1. The five feature blocks.

Definition 3.1.1 (Feature map and schema). The deterministic feature map $x: \mathcal{A} \rightarrow [0, 10]^{21}$ with schema identifier $\text{fv} \in \mathbb{Z}_{\geq 1}$ concatenates the five blocks of Definitions 3.1.2–3.1.7,

$$x(a) = (x_{\text{struct}}(a), x_{\text{resume}}(a), x_{\text{kw}}(a), x_{\text{essay}}(a), x_{\text{pen}}(a)) \in [0, 10]^3 \times [0, 10]^1 \times [0, 10]^4 \times [0, 10]^6 \times \{0, 10\}^7.$$

Definition 3.1.2 (Structured coordinates). For a whitelist $\mathcal{W}$ of course codes, the graduate codes $\mathcal{G}$ of the form MATH GR5xxx and MATH GR6xxx, and nondecreasing $\phi, \psi: [0, 1] \rightarrow [0, 1]$, for instance $\phi(u) = u^\alpha$ with $\alpha \in (0, 1]$,

$$C(a) := U(a) \cap (\mathcal{W} \cup \mathcal{G}), \quad |C(a)| \leq 6, \quad x_{\text{gpa}}(a) := 10 \phi\left(\frac{\text{GPA}(a)}{4.33}\right), \quad x_{\text{upper}}(a) := 10 \psi\left(\frac{|C(a)|}{6}\right),$$

$$x_{\text{calc}}(a) := 10 \cdot \mathbf{1}\{\text{Calc}(a) = \text{yes}\}, \quad x_{\text{struct}}(a) := (x_{\text{gpa}}, x_{\text{calc}}, x_{\text{upper}})(a) \in [0, 10] \times \{0, 10\} \times [0, 10].$$

Definition 3.1.3 (Resume length coordinate). For $\ell(a) := |T(a)|$, $L_0 > 0$, and nondecreasing $s: [0, \infty) \rightarrow [0, 1]$ with $s(0) = 0$, such as $\min\{1, u\}$ or $1 - e^{-u}$,

$$x_{\text{resume}}(a) := x_{\text{rlen}}(a) := 10 s\left(\frac{\ell(a)}{L_0}\right) \in [0, 10], \quad \ell(a) = 0 \implies x_{\text{rlen}}(a) = 0.$$

Definition 3.1.4 (Keyword coordinates). For $\mathcal{K} := \{\text{research}, \text{teach}, \text{leader}, \text{awards}\}$ and, for each $c \in \mathcal{K}$, a finite dictionary $D_c \subset \Sigma^*$ and a threshold $\tau_c \geq 1$,

$$N_c(a) := \sum_{v \in D_c} \mathbf{1}\{v \sqsubseteq T(a)\}, \quad x_c(a) := 10 \min\left\{1, \frac{N_c(a)}{\tau_c}\right\} \quad \left(\text{or } 10(1 - e^{-N_c(a)/\tau_c})\right),$$

$$x_{\text{kw}}(a) := (x_c(a))_{c \in \mathcal{K}} \in [0, 10]^4, \quad \text{KW}(a) := \frac{1}{4} \sum_{c \in \mathcal{K}} x_c(a) \in [0, 10].$$

Example 3.1.5 (Keyword saturation). For $\tau_{\text{research}} = 3$, $v := N_{\text{research}}$, $\chi := x_{\text{research}}$, and resumes containing exactly REU, preprint, poster, or only REU,

$$v(a) = 3 \implies \chi(a) = 10, \quad v(a) = 1 \implies \chi(a) = \frac{10}{3}, \quad v(a) \geq 3 \implies \chi(a) = 10.$$

Definition 3.1.6 (Essay coordinates). For word and sentence counts $\text{wc}, \text{sent} : \Sigma^* \rightarrow \mathbb{Z}_{\geq 0}$, word bounds $m \leq M$, reflection cues $R \subset \Sigma^*$ (realized, learned, mistake, revised, iterated, feedback, persist), action tokens $P \subset \Sigma^*$ (talk, problem session, reading group, outreach, workshop), a mathematical lexicon $\mathcal{M} \subset \Sigma^*$, thresholds $\rho, \varpi, \mu \geq 1$, and $\bullet \in \{A, B\}$,

$$\begin{aligned} x_{\text{band}, \bullet}(a) &:= 10 \cdot \mathbf{1}\{2 \leq \text{sent}(E_{\bullet}(a)) \leq 4\} \quad \text{or} \quad x_{\text{band}, \bullet}(a) := 10 \cdot \mathbf{1}\{m \leq \text{wc}(E_{\bullet}(a)) \leq M\}, \\ x_{\text{arc}}(a) &:= 10 \min \left\{ 1, \frac{1}{\rho} \sum_{v \in R} \mathbf{1}\{v \sqsubseteq E_A(a)\} \right\}, \quad x_{\text{plan}}(a) := 10 \min \left\{ 1, \frac{1}{\varpi} \sum_{v \in P} \mathbf{1}\{v \sqsubseteq E_B(a)\} \right\}, \\ x_{\text{math}, \bullet}(a) &:= 10 \min \left\{ 1, \frac{1}{\mu} \sum_{v \in \mathcal{M}} \mathbf{1}\{v \sqsubseteq E_{\bullet}(a)\} \right\}, \\ x_{\text{essay}}(a) &:= (x_{\text{band}, A}, x_{\text{band}, B}, x_{\text{arc}}, x_{\text{plan}}, x_{\text{math}, A}, x_{\text{math}, B})(a) \in [0, 10]^6. \end{aligned}$$

Definition 3.1.7 (Penalty block). The block $x_{\text{pen}}(a) \in \{0, 10\}^7$ lists the indicators $10 \cdot \mathbf{1}\{\text{check fails}\}$ of seven validation checks, the sign of each influence on the rank being left to the learned weight.

Example 3.1.8 (Essay decomposition). For a_1, a_2 agreeing in every coordinate except x_{plan} , with $E_B(a_1)$ naming at least ϖ actions of P and $E_B(a_2)$ none,

$$x_{\text{plan}}(a_1) = 10, \quad x_{\text{plan}}(a_2) = 0, \quad \Delta_{12}(w) = \langle w, d_{12} \rangle = 10 w_{\text{plan}}, \quad \mathbb{P}(a_1 \succ a_2 \mid w) = \sigma(10 w_{\text{plan}}).$$

Proposition 3.1.9 (Boundedness and sensitivity of the feature map). For $a, b \in \mathcal{A}$, $w \in \mathbb{R}^d$, $c \in \mathcal{K}$, and $x'_c(a)$ the keyword coordinate after inserting one phrase into $T(a)$,

$$\|x(a)\|_{\infty} \leq 10, \quad \|x(a) - x(b)\|_2^2 \leq 100d, \quad |s_w(a) - s_w(b)| \leq 10 \|w\|_1, \quad 0 \leq x'_c(a) - x_c(a) \leq \frac{10}{\tau_c},$$

the last for both saturations of Definition 3.1.4.

Proof.

$$x(a), x(b) \in [0, 10]^d \implies \|d_{ab}\|_{\infty} \leq 10 \xrightarrow{2.1.3} \|d_{ab}\|_2^2 \leq 100d, \quad |s_w(a) - s_w(b)| = |\langle w, d_{ab} \rangle| \leq 10 \|w\|_1,$$

$$N_c(a) \leq N'_c(a) \leq N_c(a) + 1 \implies 0 \leq \min \left\{ 1, \frac{N'_c(a)}{\tau_c} \right\} - \min \left\{ 1, \frac{N_c(a)}{\tau_c} \right\} \leq \frac{1}{\tau_c},$$

$$0 \leq e^{-N_c(a)/\tau_c} - e^{-N'_c(a)/\tau_c} \leq 1 - e^{-1/\tau_c} \leq \frac{1}{\tau_c}.$$

□

4. LINEAR SCORING MODEL

The score is linear so that each weight is the marginal effect of one named signal, and its geometry is that of projection onto $\hat{w} := w/\|w\|_2$ [5], [8, Chapter 3]. Throughout, $\xi \in \mathbb{R}^d$ denotes a feature vector and $H_w(c)$ the level sets of Lemma 2.1.4.

4.1. Scores, level sets, and scale.

Definition 4.1.1 (Score geometry). For $w \in \mathbb{R}^d \setminus \{0\}$ and applications a, b ,

$$s_w(a) = \|w\|_2 \langle \hat{w}, x(a) \rangle, \quad x(a) \in H_w(s_w(a)), \quad \text{dist}(H_w(s_w(a)), H_w(s_w(b))) \xrightarrow{2.1.4} \frac{|\Delta_{ab}(w)|}{\|w\|_2},$$

so $x(a)$ and $x(b)$ lie on parallel hyperplanes whose separation along $\hat{w}$ is the margin divided by $\|w\|_2$.

Proposition 4.1.2 (Scale and order). For $w \in \mathbb{R}^d \setminus \{0\}$, $c > 0$, and applications a, b ,

$$s_{cw} = c s_w, \quad \Delta_{ab}(cw) = c \Delta_{ab}(w), \quad s_w(a) < s_w(b) \iff s_{cw}(a) < s_{cw}(b) \iff \langle \hat{w}, x(a) \rangle < \langle \hat{w}, x(b) \rangle,$$

$$\frac{\partial}{\partial \xi_k} \langle w, \xi \rangle = w_k, \quad w \in [0, \infty)^d \implies 0 \leq s_w(a) \leq 10 \|w\|_1,$$

so the ordering depends on $\hat{w}$ alone while the probabilities $\sigma(\Delta_{ab}(w))$ depend on $\|w\|_2$ as well.

Proof.

$$s_{cw}(a) = \langle cw, x(a) \rangle = c s_w(a) \implies \Delta_{ab}(cw) = c \Delta_{ab}(w), \quad c > 0 \implies s_w(a) < s_w(b) \iff s_{cw}(a) < s_{cw}(b),$$

$$c := \|w\|_2^{-1} \implies s_{cw}(a) = \langle \hat{w}, x(a) \rangle, \quad \frac{\partial}{\partial \xi_k} \sum_{l=1}^d w_l \xi_l = w_k \xrightarrow{2.1.3} 0 \leq s_w(a) \leq 10 \|w\|_1.$$

□

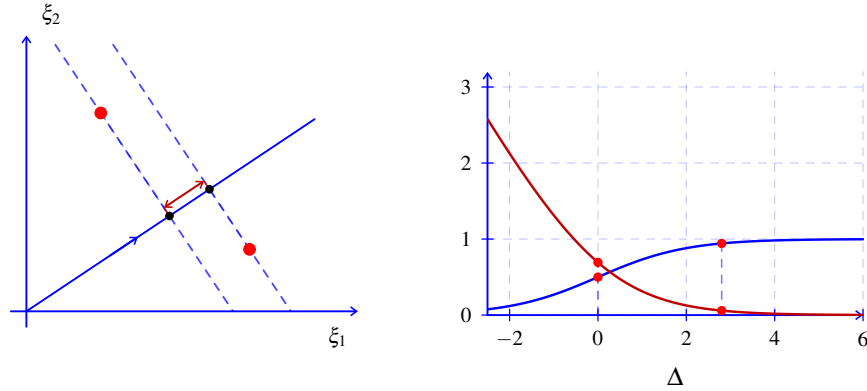


FIGURE 2. For $w = (3, 2)$, $x(a) = (1.2, 3.2)$, and $x(b) = (3.6, 1)$, the left panel shows $H_w(10)$, $H_w(12.8)$, and separation $2.8/\sqrt{13}$; the right plots σ in blue and $\ell(\Delta) = \log(1 + e^{-\Delta})$ in red, with $(\sigma, \ell)(0) = (\frac{1}{2}, \log 2)$ and $(\sigma, \ell)(2.8) \approx (.943, .059)$.

5. PAIRWISE TRAINING OBJECTIVE

Maximum likelihood for the Bradley–Terry probabilities of Definition 1.1.2 is the minimization of a sum of logistic losses of signed margins, whose ℓ_2 -regularized version is strongly convex and smooth with explicit constants, so that gradient descent converges linearly [1], [5], [8, Chapter 4], [9, Chapter 3]. Throughout, $\mathcal{D}$ is a pairwise dataset, $N := |\mathcal{D}|$, sums over (i, j, y) run over $\mathcal{D}$, σ' is evaluated at $\langle w, d_{ij} \rangle$ unless stated otherwise, and $\lambda \geq 0$.

5.1. Loss, gradient, convexity, and gradient descent.

Lemma 5.1.1 (Negative log-likelihood is the logistic loss). *For $(i, j, y) \in \mathcal{D}$, $z = 2y - 1$, and $\Delta = \Delta_{ij}(w)$,*

$$\begin{aligned} -y \log \sigma(\Delta) - (1-y) \log(1 - \sigma(\Delta)) &= \log(1 + e^{-z\Delta}) = \ell_{ij}(w), \\ \frac{1}{N} \sum_{(i,j,y)} \ell_{ij}(w) &= -\frac{1}{N} \log \prod_{(i,j,y)} \mathbb{P}(a_i \succ a_j \mid w)^y \mathbb{P}(a_j \succ a_i \mid w)^{1-y}. \end{aligned}$$

Proof.

$$\begin{aligned} y = 1 &\implies z = 1, \quad -\log \sigma(\Delta) \xrightarrow{2.1.2} \log(1 + e^{-\Delta}) = \log(1 + e^{-z\Delta}), \\ y = 0 &\implies z = -1, \quad -\log(1 - \sigma(\Delta)) \xrightarrow{2.1.2} -\log \sigma(-\Delta) = \log(1 + e^{\Delta}) = \log(1 + e^{-z\Delta}), \\ \mathbb{P}(a_j \succ a_i \mid w) &= \sigma(\Delta_{ji}(w)) = \sigma(-\Delta) \xrightarrow{2.1.2} 1 - \sigma(\Delta) \implies -\log(\sigma(\Delta)^y (1 - \sigma(\Delta))^{1-y}) = \ell_{ij}(w). \end{aligned}$$

□

Lemma 5.1.2 (Gradient and Hessian). *For $w \in \mathbb{R}^d$,*

$$\begin{aligned} \nabla \ell_{ij}(w) &= (\sigma(\Delta_{ij}(w)) - y_{ij}) d_{ij}, \quad \nabla^2 \ell_{ij}(w) = \sigma'(\Delta_{ij}(w)) d_{ij} \cdot d_{ij}^\top, \\ \nabla L_\lambda(w) &= \frac{1}{N} \sum_{(i,j,y)} (\sigma(\langle w, d_{ij} \rangle) - y) d_{ij} + \lambda w, \quad \nabla^2 L_\lambda(w) = \frac{1}{N} \sum_{(i,j,y)} \sigma'(\langle w, d_{ij} \rangle) d_{ij} \cdot d_{ij}^\top + \lambda I. \end{aligned}$$

Proof. Write $\Delta = \langle w, d \rangle$ with $d = d_{ij}$, $y = y_{ij}$, $z = 2y - 1$, and differentiate $\ell_{ij}(w) = \log(1 + e^{-z\Delta})$ by the chain rule.

$$\begin{aligned} \nabla_w \log(1 + e^{-z\Delta}) &= \frac{-ze^{-z\Delta}}{1 + e^{-z\Delta}} d = -z \sigma(-z\Delta) d, \quad \nabla_w \sigma(\langle w, d \rangle) = \sigma'(\Delta) d, \\ z = 1 &\implies -\sigma(-\Delta) d \xrightarrow{2.1.2} (\sigma(\Delta) - 1) d, \quad z = -1 \implies \sigma(\Delta) d, \quad \nabla \ell_{ij}(w) = (\sigma(\Delta) - y) d, \\ \nabla^2 \ell_{ij}(w) &= \sigma'(\Delta) d \cdot d^\top, \quad \nabla \left(\frac{\lambda}{2} \|w\|_2^2 \right) = \lambda w, \quad \nabla^2 \left(\frac{\lambda}{2} \|w\|_2^2 \right) = \lambda I. \end{aligned}$$

□

Proposition 5.1.3 (Convexity, strong convexity, and smoothness). *For $\lambda \geq 0$, $\Lambda = \lambda + \frac{1}{4N} \sum_{(i,j,y)} \|d_{ij}\|_2^2$, and all $w \in \mathbb{R}^d$,*

$$\begin{aligned} \lambda I &\preceq \nabla^2 L_\lambda(w) \preceq \Lambda I \preceq (\lambda + 25d) I, \quad L_\lambda \text{ convex}, \\ L_\lambda \text{ strictly convex} &\iff \lambda > 0 \text{ or } \text{span}\{d_{ij} : (i, j, y) \in \mathcal{D}\} = \mathbb{R}^d, \end{aligned}$$

and for $\lambda > 0$ there is a unique minimizer w_λ , characterized by its stationarity equation and bounded by

$$\lambda w_\lambda = \frac{1}{N} \sum_{(i,j,y)} (y - \sigma(\langle w_\lambda, d_{ij} \rangle)) d_{ij}, \quad \|w_\lambda\|_2 \leq \frac{10\sqrt{d}}{\lambda}.$$

Proof.

$$0 < \sigma' \leq \frac{1}{4} \xrightarrow{2.1.2, 5.1.2} \lambda I \preceq \nabla^2 L_\lambda(w) \preceq \lambda I + \frac{1}{4N} \sum_{(i,j,y)} d_{ij} \cdot d_{ij}^\top \preceq \Lambda I, \quad \|d_{ij}\|_2^2 \xrightarrow{3.1.9} 100d \implies \Lambda \leq \lambda + 25d,$$

$$v^\top \cdot \nabla^2 L_0(w) \cdot v = \frac{1}{N} \sum_{(i,j,y)} \sigma'(\langle w, d_{ij} \rangle) \langle v, d_{ij} \rangle^2 = 0 \iff \langle v, d_{ij} \rangle = 0 \quad \forall (i, j, y) \in \mathcal{D} \iff v \perp \text{span}\{d_{ij}\},$$

$$\lambda > 0 \implies L_\lambda(w) \geq \frac{\lambda}{2} \|w\|_2^2 \longrightarrow \infty \quad (\|w\|_2 \rightarrow \infty) \implies \exists! w_\lambda,$$

$$\nabla L_\lambda(w_\lambda) = 0 \xLeftrightarrow{5.1.2} \lambda w_\lambda = \frac{1}{N} \sum_{(i,j,y)} (y - \sigma(\langle w_\lambda, d_{ij} \rangle)) d_{ij},$$

$$|y - \sigma| \leq 1, \quad \|d_{ij}\|_2 \leq 10\sqrt{d} \implies \lambda \|w_\lambda\|_2 \leq \frac{1}{N} \sum_{(i,j,y)} \|d_{ij}\|_2 \leq 10\sqrt{d}.$$

□

Theorem 5.1.4 (Gradient descent, [9]). *Let $\lambda > 0$, $w^{(0)} \in \mathbb{R}^d$, and $w^{(t+1)} := w^{(t)} - \Lambda^{-1} \nabla L_\lambda(w^{(t)})$. Then for every $t \geq 0$,*

$$\begin{aligned} L_\lambda(w^{(t)}) - L_\lambda(w_\lambda) &\leq \left(1 - \frac{\lambda}{\Lambda}\right)^t (L_\lambda(w^{(0)}) - L_\lambda(w_\lambda)), \\ \|w^{(t)} - w_\lambda\|_2^2 &\leq \frac{2}{\lambda} \left(1 - \frac{\lambda}{\Lambda}\right)^t (L_\lambda(w^{(0)}) - L_\lambda(w_\lambda)). \end{aligned}$$

Proof. Write $f := L_\lambda$, $f^* := f(w_\lambda)$, $g := \nabla f(w^{(t)})$, and apply Lemma A.1.1 with $\mu = \lambda$ and $L = \Lambda$ through Proposition 5.1.3.

$$\begin{aligned} \nabla^2 f \preceq \Lambda I &\xrightarrow{\text{A.1.1}} f(w^{(t+1)}) \leq f(w^{(t)}) - \frac{1}{2\Lambda} \|g\|_2^2, \quad \lambda I \preceq \nabla^2 f \xrightarrow{\text{A.1.1}} \|g\|_2^2 \geq 2\lambda (f(w^{(t)}) - f^*), \\ f(w^{(t+1)}) - f^* &\leq \left(1 - \frac{\lambda}{\Lambda}\right) (f(w^{(t)}) - f^*) \implies f(w^{(t)}) - f^* \leq \left(1 - \frac{\lambda}{\Lambda}\right)^t (f(w^{(0)}) - f^*), \\ \lambda I \preceq \nabla^2 f &\xrightarrow{\text{A.1.1}} \frac{\lambda}{2} \|w^{(t)} - w_\lambda\|_2^2 \leq f(w^{(t)}) - f^* \leq \left(1 - \frac{\lambda}{\Lambda}\right)^t (f(w^{(0)}) - f^*). \end{aligned}$$

□

Remark 5.1.5 (Optimization in practice). At the scale $d = 21$ and N between 10 and 10^3 , a fixed number of epochs of the update $w^{(t+1)} = w^{(t)} - \eta \nabla L_\lambda(w^{(t)})$ with small λ suffices, each gradient costing $O(Nd)$ operations, since by Theorem 5.1.4

$$\Lambda \leq \lambda + 25d = \lambda + 525, \quad t \geq \frac{\Lambda}{\lambda} \log \frac{L_\lambda(w^{(0)}) - L_\lambda(w_\lambda)}{\varepsilon} \implies L_\lambda(w^{(t)}) - L_\lambda(w_\lambda) \leq \varepsilon \quad (\varepsilon > 0).$$

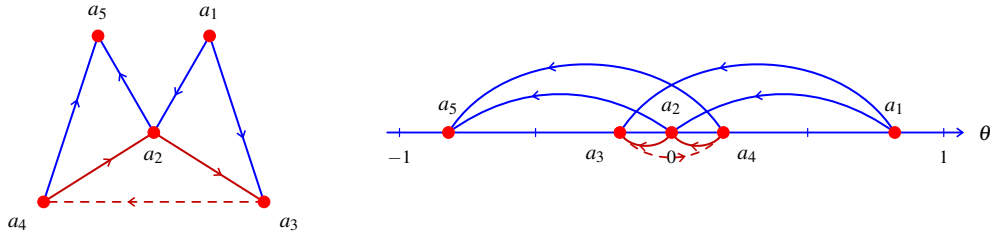


FIGURE 3. Seven comparisons on five applicants: the blue edges $a_1 \rightarrow a_2, a_1 \rightarrow a_3, a_4 \rightarrow a_5, a_2 \rightarrow a_5$ have probabilities .694, .732, .732, .694, the red cycle $a_2 \rightarrow a_3 \rightarrow a_4 \rightarrow a_2$ has probabilities .546, .408, .546 with its violated edge dashed, and the right panel places the same arrows at the $\lambda = .1$ fit $\theta = (.82, 0, -.19, .19, -.82)$.

6. CALIBRATION AND REPORTING

The objective of § 5 determines w through logit differences, so the raw scale of s_w is not intrinsic and applicant-facing reporting uses a monotone calibration to a bounded scale that preserves order [7], [8, Chapter 4]. Throughout, $\mathcal{P} = \{a_1, \dots, a_n\} \subset \mathcal{A}$ is the pool of a session or a reference set, e_c the coordinate vector of $c \in \mathcal{K}$, every submission stores the record $(s_w(a), \hat{s}(a), \text{KW}(a), x(a))$, and a session without a pool file logs $s_w(a)$ and sets the degenerate, non-comparable value $\hat{s}(a) := 10$, so that the whole pipeline of § 3–§ 5 is the composite

$$\mathcal{A} \xrightarrow{x} [0, 10]^{21} \xrightarrow{\langle w_\lambda, \cdot \rangle} \mathbb{R} \xrightarrow{10\hat{F}_{w_\lambda}} \left\{ \frac{10}{n}, \frac{20}{n}, \dots, 10 \right\}.$$

6.1. Percentile calibration and decomposed reporting.

Definition 6.1.1 (Empirical distribution and calibrated score). For $t \in \mathbb{R}$ and $a \in \mathcal{A}$,

$$\hat{F}_w(t) := \frac{1}{n} \sum_{m=1}^n \mathbf{1}\{s_w(a_m) \leq t\}, \quad \hat{s}(a) := 10\hat{F}_w(s_w(a)) = \frac{10}{n} |\{m : s_w(a_m) \leq s_w(a)\}| \in [0, 10].$$

Proposition 6.1.2 (Order preservation and invariance of the calibration). For $a = a_p$, $b = a_q$ in $\mathcal{P}$, $g : \mathbb{R} \rightarrow \mathbb{R}$ strictly increasing, $c > 0$, $\hat{F}_{g \circ s_w}$ the empirical distribution of $\{g(s_w(a_m))\}_m$, and $\hat{s}_w$ the calibrated score built from w ,

$$s_w(a) \gtrless s_w(b) \iff \hat{s}(a) \gtrless \hat{s}(b), \quad \hat{s}(\mathcal{P}) \subset \left\{ \frac{10}{n}, \dots, 10 \right\}, \quad \max_{\mathcal{P}} \hat{s} = 10,$$

$$\hat{F}_{g \circ s_w} \circ g \circ s_w = \hat{F}_w \circ s_w, \quad \hat{s}_{cw} = \hat{s}_w.$$

Proof. Write $S_p := \{m : s_w(a_m) \leq s_w(a_p)\}$, so that $\hat{s}(a_p) = 10|S_p|/n$ and $p \in S_p$.

$$s_w(a_p) < s_w(a_q) \implies S_p \subsetneq S_q, \quad s_w(a_p) = s_w(a_q) \implies S_p = S_q, \quad s_w(a_p) \gtrless s_w(a_q) \implies |S_p| \gtrless |S_q|,$$

$$1 \leq |S_p| \leq n, \quad s_w(a_p) = \max_m s_w(a_m) \implies S_p = \{1, \dots, n\}, \quad \{m : g(s_w(a_m)) \leq g(s_w(a_p))\} = S_p,$$

$$s_{cw} = c s_w \xrightarrow{4.1.2} \hat{F}_{cw}(s_{cw}(a_p)) = \hat{F}_{cw}(c s_w(a_p)) = \hat{F}_w(s_w(a_p)).$$

□

Remark 6.1.3 (Keyword-only summary). The diagnostic KW of Definition 3.1.4 is the fixed linear functional

$$\text{KW}(a) = \langle u, x(a) \rangle \in [0, 10], \quad u := \frac{1}{4} \sum_{c \in \mathcal{K}} e_c, \quad s_w(a) - \text{KW}(a) = \langle w - u, x(a) \rangle,$$

independent of the learned w and blind to the structured, essay, and penalty blocks, a reviewer-facing meter of resume evidence and not a replacement for s_w .

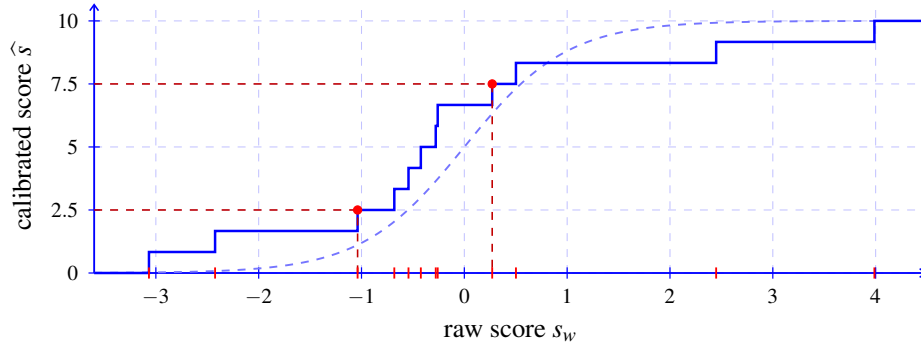


FIGURE 4. Percentile calibration for $n = 12$: the solid staircase $\hat{s} = 10\hat{F}_w(s_w)$ jumps by $10/12$ at the red score marks, the dashed curve is $10\sigma(2s)$, and the red guides give $\hat{s}(b) = 2.5$ at $s_w(b) = -1.038$ and $\hat{s}(a) = 7.5$ at $s_w(a) = 0.271$.

7. PROOF OF THEOREM A

Theorem 7.0.1 (Regularized pairwise logistic ranking). Let $x : \mathcal{A} \rightarrow [0, 10]^d$ be a feature map, $\mathcal{D}$ a pairwise dataset, $\lambda > 0$, $\Lambda = \lambda + \frac{1}{4N} \sum_{(i,j,y)} \|d_{ij}\|_2^2$, and $\mathcal{P}$ a pool. Then

- (1) $\lambda I \preceq \nabla^2 L_\lambda(w) \preceq \Lambda I \preceq (\lambda + 25d)I$ for all $w \in \mathbb{R}^d$, and L_λ has a unique minimizer w_λ with $\|w_\lambda\|_2 \leq 10\sqrt{d}/\lambda$.

Proof.

$$0 < \sigma' \leq \frac{1}{4} \xrightarrow{2.1.2, 3.1.9} \|d_{ij}\|_2^2 \leq 100d \xrightarrow{5.1.3} \lambda I \preceq \nabla^2 L_\lambda \preceq \Lambda I \preceq (\lambda + 25d)I, \quad \exists! w_\lambda, \quad \|w_\lambda\|_2 \leq \frac{10\sqrt{d}}{\lambda},$$

□

(2) w_λ is the unique solution of $\lambda w = \frac{1}{N} \sum_{(i,j,y)} (y - \sigma(\langle w, d_{ij} \rangle)) d_{ij}$.

Proof.

$$\nabla L_\lambda(w) = 0 \xLeftrightarrow{5.1.2} \lambda w = \frac{1}{N} \sum_{(i,j,y)} (y - \sigma(\langle w, d_{ij} \rangle)) d_{ij} \xLeftrightarrow{5.1.3} w = w_\lambda,$$

□

(3) gradient descent with step Λ^{-1} satisfies $L_\lambda(w^{(t)}) - L_\lambda(w_\lambda) \leq (1 - \lambda/\Lambda)^t (L_\lambda(w^{(0)}) - L_\lambda(w_\lambda))$.

Proof.

$$w^{(t+1)} = w^{(t)} - \Lambda^{-1} \nabla L_\lambda(w^{(t)}) \xrightarrow{5.1.4} L_\lambda(w^{(t)}) - L_\lambda(w_\lambda) \leq \left(1 - \frac{\lambda}{\Lambda}\right)^t (L_\lambda(w^{(0)}) - L_\lambda(w_\lambda)),$$

□

(4) for $a, b \in \mathcal{P}$, $s_w(a) \leq s_w(b) \iff \widehat{s}(a) \leq \widehat{s}(b)$, and $\widehat{s}$ is invariant under $w \mapsto cw$ for $c > 0$.

Proof.

$$a, b \in \mathcal{P} \xrightarrow{6.1.2} (s_w(a) \leq s_w(b) \iff \widehat{s}(a) \leq \widehat{s}(b)), \quad c > 0 \xrightarrow{6.1.2} \widehat{s}_{cw} = \widehat{s}_w.$$

□

8. THE CONJECTURE

The regularization path $\lambda \mapsto w_\lambda$ is the one part of the pipeline whose qualitative behaviour is not settled by § 5–§ 6, and its large- λ limit is the anchor for what follows [8, Chapter 3], [7]. Throughout, $\mathcal{P}$ is a pool, $\mathcal{D}$ a pairwise dataset, $w^* \in \mathbb{R}^d$ a true weight vector, and

$$\bar{d} := \frac{1}{N} \sum_{(i,j,y) \in \mathcal{D}} (y_{ij} - \tfrac{1}{2}) d_{ij}.$$

8.1. The regularization path and its limit.

Proposition 8.1.1 (Large regularization limit). *For $\lambda > 0$,*

$$\|\lambda w_\lambda - \bar{d}\|_2 \leq (\Lambda - \lambda) \|w_\lambda\|_2 \leq \frac{250d^{3/2}}{\lambda}, \quad \lim_{\lambda \rightarrow \infty} \lambda w_\lambda = \bar{d}.$$

Proof.

$$\nabla L_\lambda(w_\lambda) = 0 \xLeftrightarrow{5.1.2} \lambda w_\lambda - \bar{d} = \frac{1}{N} \sum_{(i,j,y)} \left(\tfrac{1}{2} - \sigma(\langle w_\lambda, d_{ij} \rangle)\right) d_{ij}, \quad \sigma(0) = \tfrac{1}{2}, \quad \sigma' \leq \tfrac{1}{4} \xrightarrow{2.1.2} \left|\tfrac{1}{2} - \sigma(t)\right| \leq \tfrac{|t|}{4},$$

$$\|\lambda w_\lambda - \bar{d}\|_2 \leq \frac{1}{4N} \sum_{(i,j,y)} |\langle w_\lambda, d_{ij} \rangle| \|d_{ij}\|_2 \leq \frac{\|w_\lambda\|_2}{4N} \sum_{(i,j,y)} \|d_{ij}\|_2^2 = (\Lambda - \lambda) \|w_\lambda\|_2 \xrightarrow{3.1.9, 5.1.3} \frac{25d \cdot 10\sqrt{d}}{\lambda}.$$

□

Conjecture 8.1.2 (Finiteness and stabilization of the path). *For $a, b \in \mathcal{P}$ with $\Delta_{ab}(w_\lambda) \not\equiv 0$, and some $\lambda_0 = \lambda_0(\mathcal{P}, \mathcal{D})$,*

$$|\{\lambda > 0 : \Delta_{ab}(w_\lambda) = 0\}| < \infty, \quad \lambda \geq \lambda_0 \implies (\Delta_{ab}(w_\lambda) < 0 \iff \langle \bar{d}, x(a) - x(b) \rangle < 0).$$

Conjecture 8.1.3 (Sample complexity of the calibrated order). *For labels drawn independently from Remark 2.1.5 with weights w^* on N pairs uniform in $\mathcal{P}$, $|\mathcal{P}| = n$, $\gamma > 0$, and $\lambda \asymp N^{-1/2}$, there are $C, c > 0$ with*

$$N \geq Cd \gamma^{-2} \log n \implies \mathbb{P}(E_\gamma) \geq 1 - n^{-c}, \quad E_\gamma := \bigcap_{|\Delta_{ab}(w^*)| \geq \gamma} \{s_{w^*}(a) < s_{w^*}(b) \iff \widehat{s}(a) < \widehat{s}(b)\}.$$

Conjecture 8.1.4 (Calibration under pool drift). *For pools $\mathcal{P}_0, \mathcal{P}_t$ of sizes n_0, n_t drawn from μ_0, μ_t on $\mathcal{A}$ with $W_1(\mu_0 \circ s_w^{-1}, \mu_t \circ s_w^{-1}) \leq \eta$, and $\widehat{s}^\mathcal{P}$ the calibration of Definition 6.1.1 against $\mathcal{P}$, determine the least $\varepsilon = \varepsilon(\eta, n_0, n_t, \delta)$ and the reference size n_0 with*

$$\mathbb{P}\left(\sup_{a \in \mathcal{A}} |\widehat{s}^{\mathcal{P}_t}(a) - \widehat{s}^{\mathcal{P}_0}(a)| \leq \varepsilon\right) \geq 1 - \delta,$$

replacing the degenerate solo value $\widehat{s}(a) = 10$ of § 6 by a score comparable across sessions.

Part 2. Appendix

APPENDIX A. FOUNDATOIONAL TOOLS

A.1. Quadratic bounds and maximum likelihood.

Lemma A.1.1 (Quadratic bounds from Hessian bounds). *Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be twice continuously differentiable with $\mu I \preceq \nabla^2 f(w) \preceq LI$ for all w , where $0 < \mu \leq L$, and let w_* be its minimizer. Then for all $v, w \in \mathbb{R}^d$,*

$$\begin{aligned} f(w) + \langle \nabla f(w), v - w \rangle + \frac{\mu}{2} \|v - w\|_2^2 &\leq f(v) \leq f(w) + \langle \nabla f(w), v - w \rangle + \frac{L}{2} \|v - w\|_2^2, \\ \frac{\mu}{2} \|w - w_*\|_2^2 &\leq f(w) - f(w_*) \leq \frac{1}{2\mu} \|\nabla f(w)\|_2^2, \quad f\left(w - \frac{1}{L} \nabla f(w)\right) \leq f(w) - \frac{1}{2L} \|\nabla f(w)\|_2^2. \end{aligned}$$

Proof. Taylor's formula with integral remainder along the segment from w to v gives the first line, and the second follows by choosing v [9, Chapter 3].

$$\begin{aligned} f(v) &= f(w) + \langle \nabla f(w), v - w \rangle + \int_0^1 (1 - \theta) (v - w)^\top \cdot \nabla^2 f(w + \theta(v - w)) \cdot (v - w) d\theta, \\ \int_0^1 (1 - \theta) d\theta &= \frac{1}{2}, \quad \mu \|v - w\|_2^2 \leq (v - w)^\top \cdot \nabla^2 f(w + \theta(v - w)) \cdot (v - w) \leq L \|v - w\|_2^2, \\ v := w - \frac{1}{L} \nabla f(w) &\implies f(v) \leq f(w) - \frac{1}{L} \|\nabla f(w)\|_2^2 + \frac{L}{2} \cdot \frac{1}{L^2} \|\nabla f(w)\|_2^2 = f(w) - \frac{1}{2L} \|\nabla f(w)\|_2^2, \\ v := w_* &\implies f(w_*) \geq f(w) + \min_{u \in \mathbb{R}^d} \left(\langle \nabla f(w), u \rangle + \frac{\mu}{2} \|u\|_2^2 \right) = f(w) - \frac{1}{2\mu} \|\nabla f(w)\|_2^2, \\ w := w_*, \nabla f(w_*) = 0 &\implies f(v) \geq f(w_*) + \frac{\mu}{2} \|v - w_*\|_2^2 \quad (v \in \mathbb{R}^d). \end{aligned}$$

□

Remark A.1.2 (Maximum likelihood and ridge penalty). For conditionally independent labels with $\mathbb{P}(y_{ij} = 1 \mid w) = \sigma(\Delta_{ij}(w))$ and the Gaussian prior $w \sim \mathcal{N}(0, (N\lambda)^{-1}I)$,

$$\begin{aligned} -\frac{1}{N} \log \prod_{(i,j,y)} \sigma(\Delta_{ij}(w))^y (1 - \sigma(\Delta_{ij}(w)))^{1-y} &\xrightarrow{5.1.1} \frac{1}{N} \sum_{(i,j,y)} \ell_{ij}(w), \\ -\frac{1}{N} \log \exp\left(-\frac{N\lambda}{2} \|w\|_2^2\right) &= \frac{\lambda}{2} \|w\|_2^2, \quad L_\lambda(w) = \frac{1}{N} \sum_{(i,j,y)} \ell_{ij}(w) + \frac{\lambda}{2} \|w\|_2^2, \end{aligned}$$

so w_λ is the maximum-a-posteriori estimate, and w_0 , when $\text{span}\{d_{ij}\} = \mathbb{R}^d$ and the minimum of L_0 is attained, is the maximum-likelihood estimate [8, Chapters 3–4].

APPENDIX B. DATA ARTIFACTS

Three files connect deployed scoring to local training, and the determinism of Definition 3.1.1 is what makes a score computed in the browser reproducible from the stored records.

B.1. Records, pairs, and the model file.

Definition B.1.1 (Artifacts). For a pool $\mathcal{P} = \{a_1, \dots, a_n\}$, a pairwise dataset $\mathcal{D}$, and a weight vector w_λ trained on the schema `fv`,

$$\begin{aligned} \text{applicants.jsonl} &:= \left\{ (m, \text{fv}, x(a_m), s_{w_\lambda}(a_m), \hat{s}(a_m), \text{KW}(a_m)) \right\}_{m=1}^n, \\ \text{pairs.csv} &:= \{(i, j, y_{ij})\}_{(i,j,y) \in \mathcal{D}}, \quad \text{rank_model.json} := (\text{fv}, \lambda, w_\lambda), \end{aligned}$$

one record per applicant, one row per comparison, and one weight vector with the schema it was trained on.

Proposition B.1.2 (Reproducibility of deployed scores). *If the schema of `rank_model.json` equals the schema of the deployed feature map, then for every $a \in \mathcal{P}$ the raw score recomputed at deployment equals the stored one, and the calibrated score and the keyword summary are functions of the stored records alone,*

$$\begin{aligned} \text{fv}_{\text{model}} = \text{fv}_{\text{deploy}} &\implies \langle w_\lambda, x(a) \rangle = s_{w_\lambda}(a), & \text{KW}(a) &= \langle u, x(a) \rangle, \\ \hat{s}(a) &= \frac{10}{n} |\{m : s_{w_\lambda}(a_m) \leq s_{w_\lambda}(a)\}|. \end{aligned}$$

Proof. Definition 3.1.1, Definition 6.1.1, Remark 6.1.3. □

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